EXP NO:9 Experiment to understand Matplotlib library use cases in Data Science through visualization

AIM:
To use the Restaurant Tips data set to understand the Matplotlib library through visualization.

PROGRAM:

python

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

# Read dataset
data = pd.read_csv(
    'https://raw.githubusercontent.com/mwaskom/seaborn-data/master/tips.csv'
)

# Display dataset
data

# Display information
data.info()

# Statistical description
data.describe()

# Count values of day
data.value_counts('day')

# Countplot
sns.countplot(x='day', data=data)
plt.show()

# One-hot encoding
FinalDataset = pd.concat([
    pd.get_dummies(data['day']),
    data[['total_bill', 'tip', 'size']]
], axis=1)

FinalDataset.head()

# Scatter plot - Total Bill vs Tip
sns.scatterplot(x='total_bill', y='tip', hue='day', data=data)
plt.xlabel('Total Bill')
plt.ylabel('Tip')
plt.show()

# Scatter plot - Total Bill vs Size
sns.scatterplot(x='total_bill', y='size', hue='day', data=data)
plt.xlabel('Total Bill')
plt.ylabel('Number of People')
plt.show()

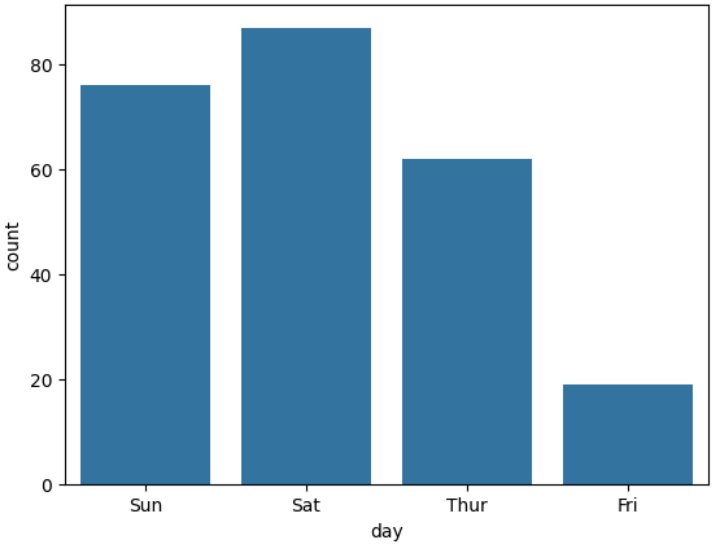
# Pairplot
sns.pairplot(data[['total_bill', 'tip', 'size', 'day']], hue='day', height=3)
plt.show()

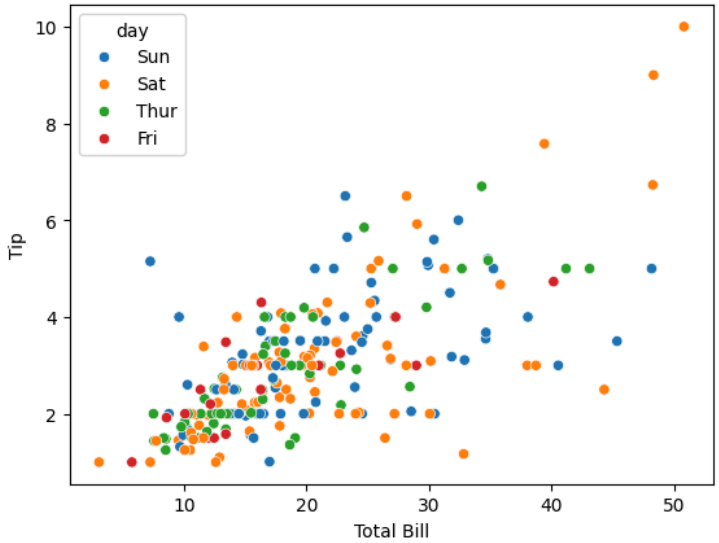
# FacetGrid - Total Bill distribution
sns.FacetGrid(data, hue='day', height=5).map(sns.histplot, 'total_bill').add_legend()
plt.show()

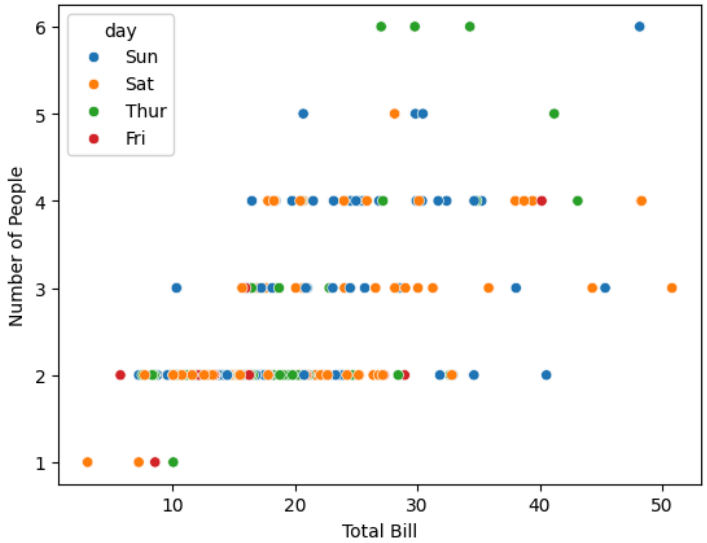
# FacetGrid - Tip distribution
sns.FacetGrid(data, hue='day', height=5).map(sns.histplot, 'tip').add_legend()
plt.show()

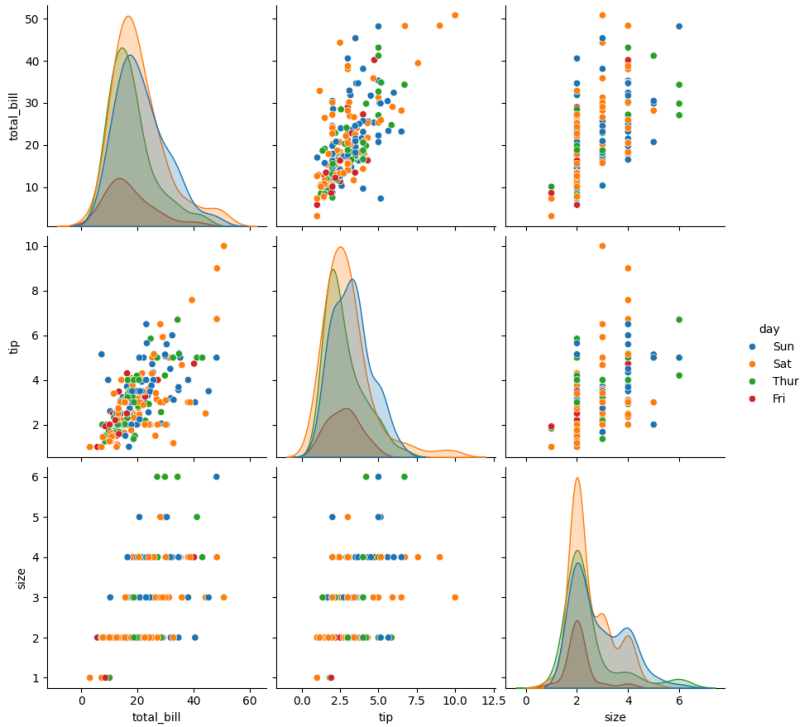
# FacetGrid - Size distribution
sns.FacetGrid(data, hue='day', height=5).map(sns.histplot, 'size').add_legend()
plt.show()
```

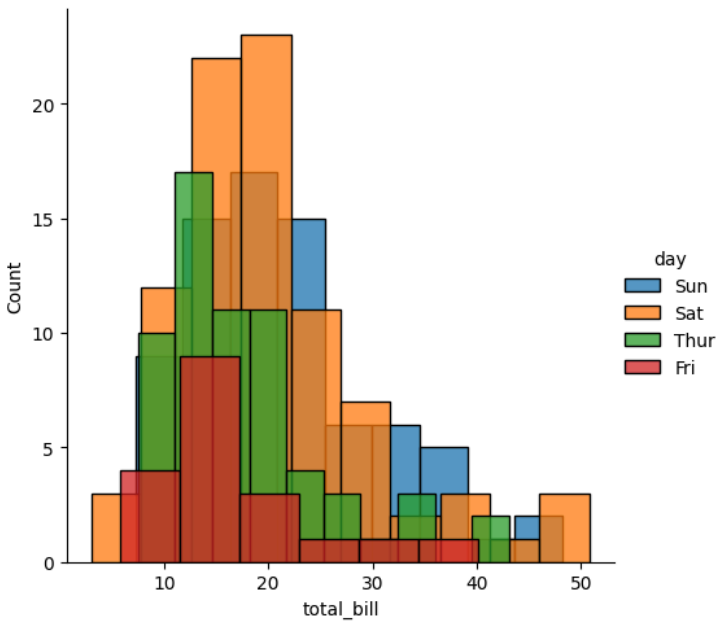
OUTPUT:

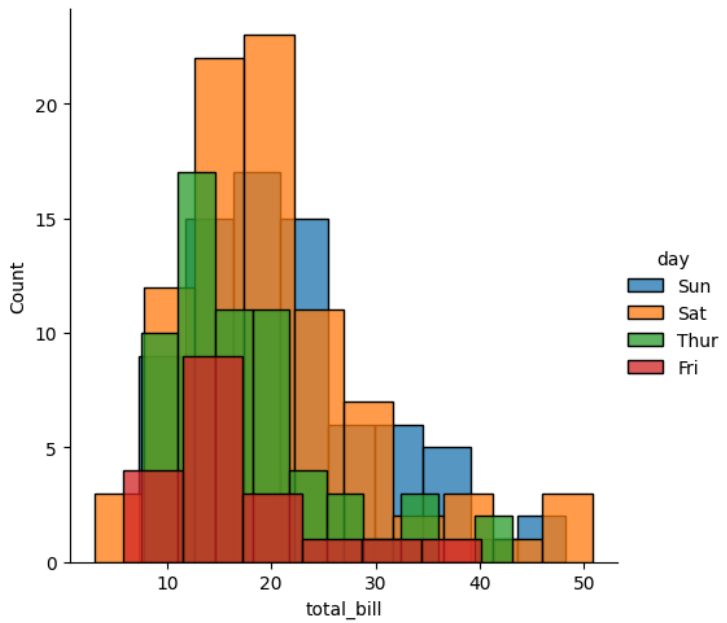

count
0
20
40
60
80
Sun
Sat
Thur
Fri
day

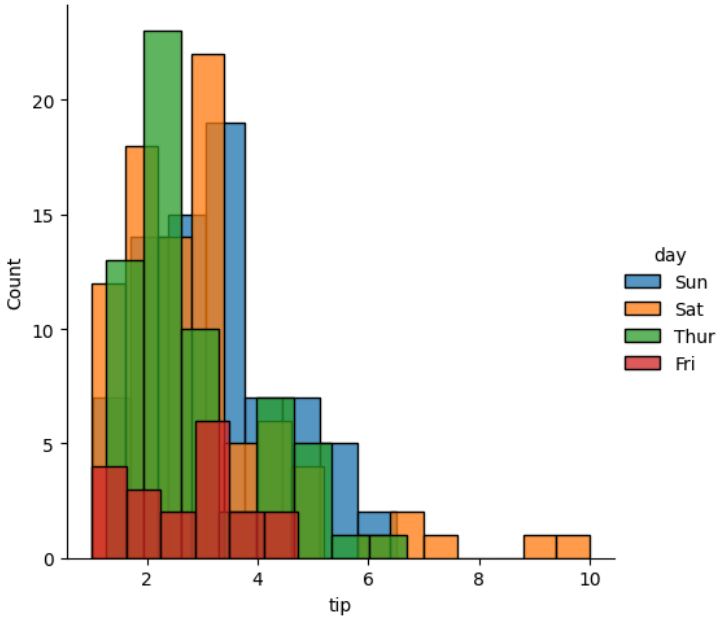

day
Sun
Sat
Thur
Fri
Tip
2
4
6
8
10
10
20
30
40
50
Total Bill

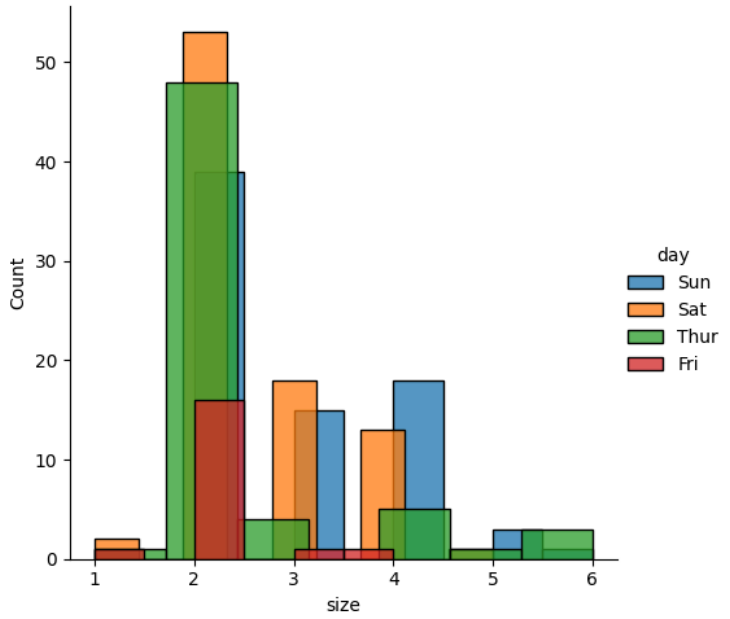

day
Sun
Sat
Thur
Fri
Number of People
Total Bill
1
2
3
4
5
6
10
20
30
40
50


total_bill
tip
size
day
Sun
Sat
Thur
Fri
0
20
40
60
0.0
2.5
5.0
7.5
10.0
12.5
0
2
4
6


Count
total_bill
day
Sun
Sat
Thur
Fri
0
5
10
15
20
10
20
30
40
50


Count
total_bill
day
Sun
Sat
Thur
Fri
0
5
10
15
20
10
20
30
40
50





RESULT:

Thus the program to understand the Matplotlib library use cases in Data Science through visualization using the Restaurant Tips data set was executed successfully.